

UNIT 5

UNIT V

CLOUD COMPUTING

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Definition of Cloud Computing – Characteristics of Cloud – Cloud Deployment Models – Cloud Service Models – Driving Factors and Challenges of Cloud – Virtualization – Load Balancing – Scalability and Elasticity – Replication – Monitoring – Cloud Services and Platforms: Compute Services – Storage Services – Application Services

Cloud Concepts & Technologies

Virtualization

Load Balancing

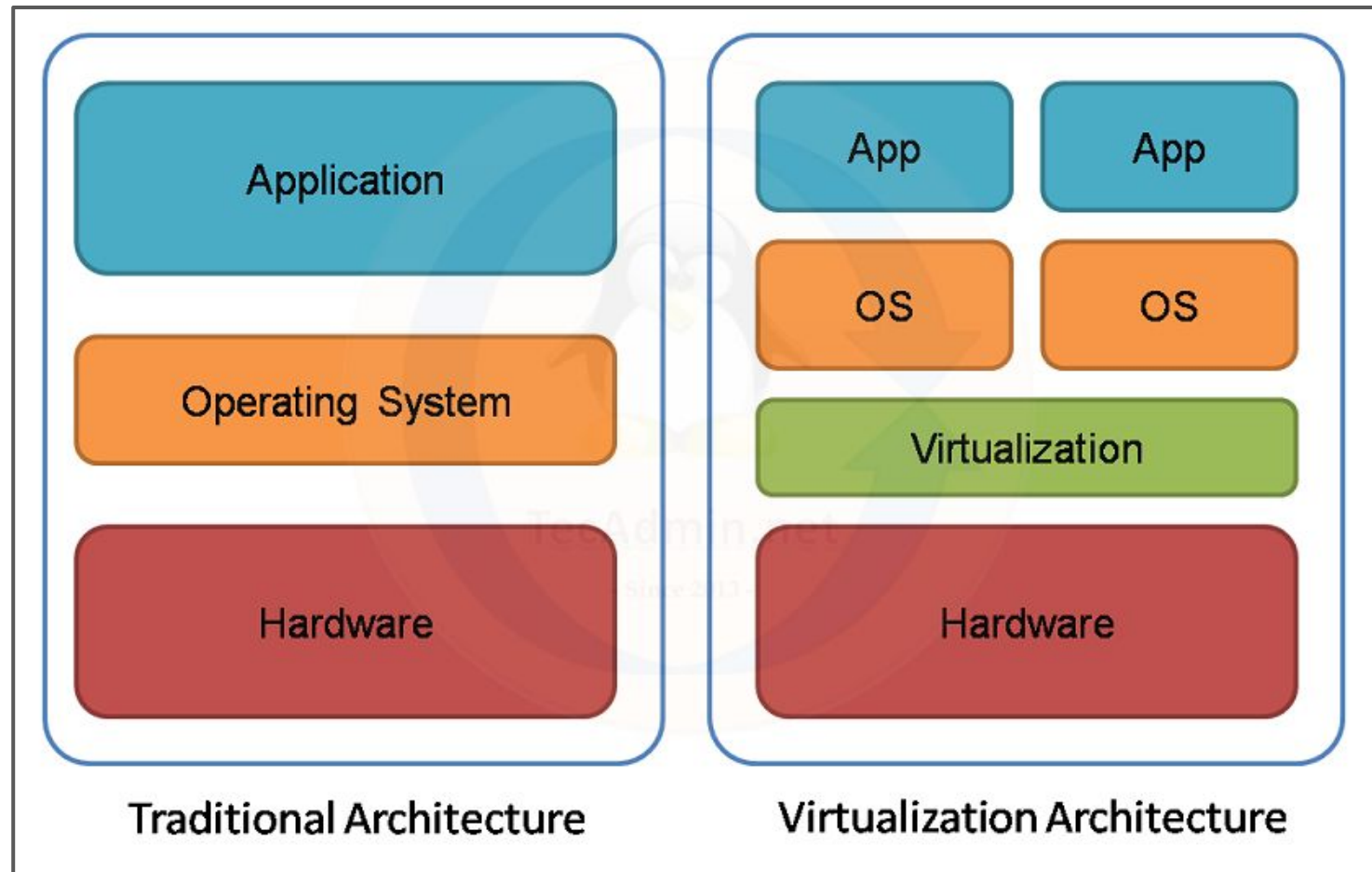
Scalability & Elasticity

Deployment

Replication

Monitoring

Virtualization



Virtualization

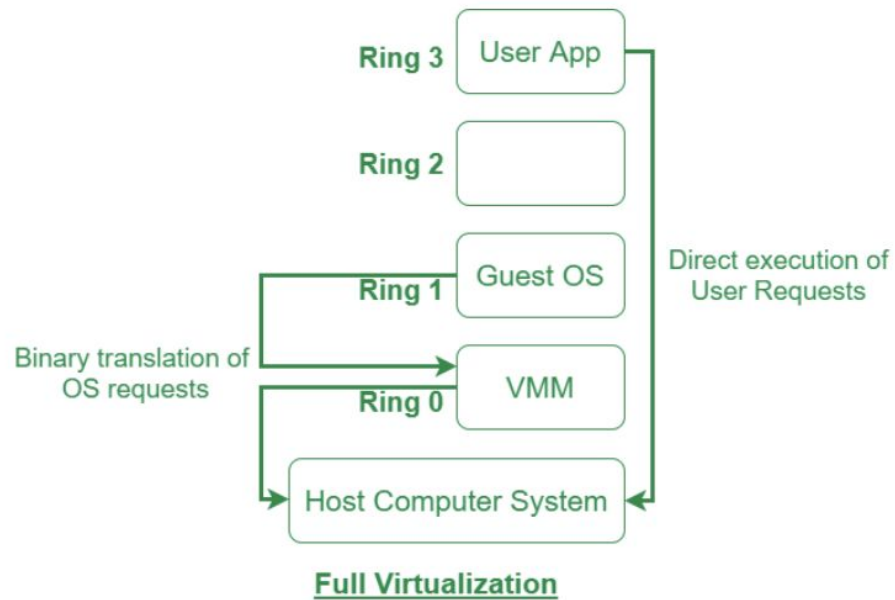
- **Virtualization** is a technique how to separate a service from the underlying physical delivery of that service. It is the process of creating a virtual version of something like computer hardware.
- It was initially developed during the mainframe era. It involves using specialized software to create a virtual or software-created version of a computing resource rather than the actual version of the same resource.
- With the help of Virtualization, multiple operating systems and applications can run on the same machine and its same hardware at the same time, increasing the utilization and flexibility of hardware.

Hypervisor

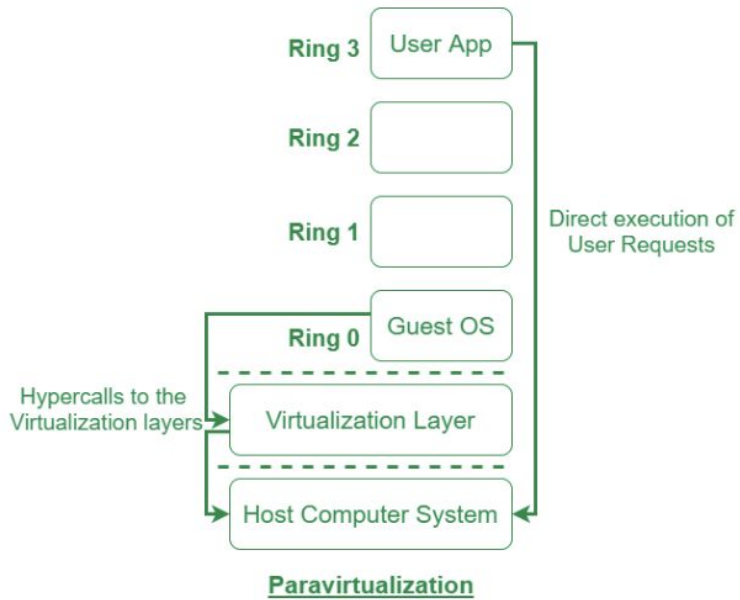
- A hypervisor, also known as a virtual machine monitor or VMM, is software that creates and runs virtual machines (VMs).
- A hypervisor allows one host computer to support multiple guest VMs by virtually sharing its resources, such as memory and processing.
- **Type 1 and Type 2**
 - The type 1 hypervisor sits on top of the bare metal server and has direct access to the hardware resources. Because of this, the type 1 hypervisor is also known as a *bare metal hypervisor*.
 - In contrast, the type 2 hypervisor is an application installed on the host operating system.

	Type 1 hypervisor	Type 2 hypervisor
Also known as	Bare metal hypervisor.	Hosted hypervisor.
Runs on	Underlying physical host machine hardware.	Underlying operating system (host OS).
Best suited for	Large, resource-intensive, or fixed-use workloads.	Desktop and development environments.
Can it negotiate dedicated resources?	Yes.	No.
Knowledge required	System administrator-level knowledge.	Basic user knowledge.
Examples	VMware ESXi, Microsoft Hyper-V, KVM.	Oracle VM VirtualBox, VMware Workstation, Microsoft Virtual PC.

1. Full Virtualization: Full Virtualization was introduced by IBM in the year 1966. It is the first software solution for server virtualization and uses binary translation and direct approach techniques. In full virtualization, guest OS is completely isolated by the virtual machine from the virtualization layer and hardware. Microsoft and Parallels systems are examples of full virtualization.



2. Paravirtualization: Paravirtualization is the category of CPU virtualization which uses hypercalls for operations to handle instructions at compile time. In paravirtualization, guest OS is not completely isolated but it is partially isolated by the virtual machine from the virtualization layer and hardware. VMware and Xen are some examples of paravirtualization.



S.No.	Full Virtualization	Paravirtualization
1.	In Full virtualization, virtual machines permit the execution of the instructions with the running of unmodified OS in an entirely isolated way.	In paravirtualization, a virtual machine does not implement full isolation of OS but rather provides a different API which is utilized when OS is subjected to alteration.
2.	Full Virtualization is less secure.	While the Paravirtualization is more secure than the Full Virtualization.
3.	Full Virtualization uses binary translation and a direct approach as a technique for operations.	While Paravirtualization uses hypercalls at compile time for operations.
4.	Full Virtualization is slow than paravirtualization in operation.	Paravirtualization is faster in operation as compared to full virtualization.
5.	Full Virtualization is more portable and compatible.	Paravirtualization is less portable and compatible.
6.	Examples of full virtualization are Microsoft and Parallels systems.	Examples of paravirtualization are Microsoft Hyper-V, Citrix Xen, etc.
7.	It supports all guest operating systems without modification.	The guest operating system has to be modified and only a few operating systems support it.

Load Balancing

- Load balancing is an essential technique used in cloud computing to optimize resource utilization and ensure that no single resource is overburdened with traffic.
- It is a process of distributing workloads across multiple computing resources, such as servers, virtual machines, or containers, to achieve better performance, availability, and scalability.

1. In cloud computing, load balancing can be implemented at various levels, including the network layer, application layer, and database layer. The most common load balancing techniques used in cloud computing are:
2. Network Load Balancing: This technique is used to balance the network traffic across multiple servers or instances. It is implemented at the network layer and ensures that the incoming traffic is distributed evenly across the available servers.
3. Application Load Balancing: This technique is used to balance the workload across multiple instances of an application. It is implemented at the application layer and ensures that each instance receives an equal share of the incoming requests.
4. Database Load Balancing: This technique is used to balance the workload across multiple database servers. It is implemented at the database layer and ensures that the incoming queries are distributed evenly across the available database servers.

Round Robin

In round robin load balancing, the servers are selected one by one to serve the incoming requests in a non-hierarchical circular fashion with no priority assigned to a specific server.

Weighted Round Robin

In weighted round robin load balancing, servers are assigned some weights. The incoming requests are proportionally routed using a static or dynamic ratio of respective weights.

Low Latency

In low latency load balancing the load balancer monitors the latency of each server. Each incoming request is routed to the server which has the lowest latency.

Least Connections

In least connections load balancing, the incoming requests are routed to the server with the least number of connections.

Priority

In priority load balancing, each server is assigned a priority. The incoming traffic is routed to the highest priority server as long as the server is available. When the highest priority server fails, the incoming traffic is routed to a server with a lower priority.

Overflow

Overflow load balancing is similar to priority load balancing. When the incoming requests to highest priority server overflow, the requests are routed to a lower priority server.

Sticky sessions

In this approach all the requests belonging to a user session are routed to the same server. These sessions are called sticky sessions. The benefit of this approach is that it makes session management simple. However, a drawback of this approach is that if a server fails all the sessions belonging to that server are lost, since there is no automatic failover possible.

Session Database

In this approach, all the session information is stored externally in a separate session database, which is often replicated to avoid a single point of failure. Though, this approach involves additional overhead of storing the session information, however, unlike the sticky session approach, this approach allows automatic failover.

Browser cookies

In this approach, the session information is stored on the client side in the form of browser cookies. The benefit of this approach is that it makes the session management easy and has the least amount of overhead for the load balancer.

URL re-writing

In this approach, a URL re-write engine stores the session information by modifying the URLs on the client side. Though this approach avoids overhead on the load balancer, a drawback is that the amount of session information that can be stored is limited. For applications that require larger amounts of session information, this approach does not work.

Scalability and Elasticity

- Elasticity refers to the ability of a cloud to automatically expand or compress the infrastructural resources on a sudden up and down in the requirement so that the workload can be managed efficiently.
- This elasticity helps to minimize infrastructural costs.
- This is not applicable for all kinds of environments, it is helpful to address only those scenarios where the resource requirements fluctuate up and down suddenly for a specific time interval.
- It is not quite practical to use where persistent resource infrastructure is required to handle the heavy workload.

	Cloud Elasticity	Cloud Scalability
1	Elasticity is used just to meet the sudden up and down in the workload for a small period of time.	Scalability is used to meet the static increase in the workload.
2	Elasticity is used to meet dynamic changes, where the resources need can increase or decrease.	Scalability is always used to address the increase in workload in an organization.
3	Elasticity is commonly used by small companies whose workload and demand increases only for a specific period of time.	Scalability is used by giant companies whose customer circle persistently grows in order to do the operations efficiently.
4	It is a short term planning and adopted just to deal with an unexpected increase in demand or seasonal demands.	Scalability is a long term planning and adopted just to deal with an expected increase in demand.

Deployment Design

In this step the application deployment is created with various tiers as specified in the deployment configuration. The variables in this step include the number of servers in each tier, computing, memory and storage capacities of servers, server interconnection, resources as specified in the deployment configuration. The process of resource provisioning and deployment creation is often automated and involves a number of steps such as launching of server instances, configuration of servers, and deployment of various tiers of the application on the servers.

Performance Evaluation

Once the application is deployed in the cloud, the next step in the deployment lifecycle is to verify whether the application meets the performance requirements with the deployment. This step involves monitoring the workload on the application and measuring various workload parameters such as response time and throughput. In addition to this, the utilization of servers (CPU, memory, disk, I/O, etc.) in each tier is also monitored.

Deployment Refinement

After evaluating the performance of the application, deployments are refined so that the application can meet the performance requirements. Various alternatives can exist in this step such as vertical scaling (or scaling up), horizontal scaling (or scaling out), alternative server interconnections, alternative load balancing and replication strategies, for instance.

Replication - Copy of data @ multiple places

- The technique of storing data at more than one location or node is known as data replication.
- It is beneficial in terms of increasing data availability.
- It transfers data from a database from one server to another so that all users may share the same data without inconsistencies.
- Consequently, users may access data relevant to their jobs without interfering with the work of others in a distributed database.
- Data replication entails the continuing duplicate of transactions such that the replicate is consistently updated and synced with the source.
- However, data is available in several locations with data replication, but a specific relation must reside in just one location.

Array-based Replication

Array-based replication uses compatible storage arrays to automatically copy data from a local storage array to a remote storage array. Arrays replicate data at the disk sub-system level, therefore the type of hosts accessing the data and the type of data is not important. Thus array-based replication can work in heterogeneous environments with different operating systems. Array-based replication uses Network Attached Storage (NAS) or Storage Area Network (SAN), to replicate. A drawback of this array-based replication is that it requires similar arrays at local and remote locations. Thus the costs for setting up array-based replication are higher than the other approaches.

Network-based Replication

Network-based replication uses an appliance that sits on the network and intercepts packets that are sent from hosts and storage arrays. The intercepted packets are replicated to a secondary location. The benefits of this approach is that it supports heterogeneous environments and requires a single point of management. However, this approach involves higher initial costs due to replication hardware and software.

Host-based Replication

Host-based replication runs on standard servers and uses software to transfer data from a local to remote location. The host acts the replication control mechanism. An agent is installed on the hosts that communicates with the agents on the other hosts. Host-based replication can either be block-based or file-based. Block-based replication typically require dedicated volumes of the same size on both the local and remote servers. File-based replication requires less storage as compared to block-based storage. File-based replication gives additional allows the administrators to choose the files or folders to be replicated. Host-based replication with cloud-infrastructure provides affordable replication solutions. With host-based replication, entire virtual machines can be replicated in real-time.

Cloud Monitoring

- Monitor services provided by the cloud service provider.
- Collect and analyze data on various metrics.
- Can use pre-defined or user-defined metrics
- Also provides various statistics over data.

Key Aspects of Cloud Monitoring	Description
Performance Optimization	- Monitors metrics like CPU usage, memory, and network latency. - Identifies and addresses performance bottlenecks for efficient resource allocation.
Availability and Reliability	- Tracks server uptime, network availability, and overall infrastructure health. - Enables rapid incident detection and resolution to minimize downtime.
Incident Detection and Resolution	- Provides real-time alerts for breached thresholds or anomalies. - Facilitates quick identification and response to incidents.
Scalability Planning	- Analyzes resource utilization patterns to plan for additional capacity. - Supports dynamic scaling based on demand.
Cost Management	- Offers insights into resource usage to optimize infrastructure and control costs. - Identifies underutilized resources for efficient spending.
Security and Compliance	- Detects and responds to security incidents, ensuring compliance with regulations. - Enhances overall security posture in the cloud environment.
User Experience Management	- Monitors application response times and user interactions. - Improves the overall user experience for customer satisfaction.



THANK YOU!